

PART IV

Environmental Influences on Performance

In previous sections of this book, we discussed the physiological adjustments and coordination of systems (muscular, neural, cardiovascular, respiratory) that allow us to perform physical activity. We also saw how these systems adapt when exposed to the repeated stress of training. In [part IV](#), we turn our attention to how the body responds and adapts when challenged to exercise under extreme environmental conditions. In [chapter 12](#), Exercise in Hot and Cold Environments, we examine the mechanisms by which the body regulates its internal temperature at rest and during exercise. Then we consider how the body responds and adapts to exercise in the heat and cold, along with the health risks associated with physical activity in hot and cold environments. In [chapter 13](#), Exercise at Altitude, we discuss the unique challenges that the body faces when performing physical activity under conditions of reduced atmospheric pressure (altitude) and how the body adapts to time spent at altitude. We then discuss the best way to prepare for competing at altitude and whether altitude training might help people perform better at sea level. Finally, health risks associated with ascent to high altitude are discussed.





12

Exercise in Hot and Cold Environments

In this chapter and in the web study guide

Body Temperature Regulation

Metabolic Heat Production

Transfer of Body Heat to and From the Environment

Thermoregulatory Control

ANIMATION FOR FIGURE 12.5 shows the hypothalamus' response to changes in body temperature.

ACTIVITY 12.1 *Control of Heat Exchange* explores the role of the hypothalamus in controlling body temperature.

Physiological Responses to Exercise in the Heat

Cardiovascular Function

What Limits Exercise in the Heat?

Body Fluid Balance: Sweating

VIDEO 12.1 presents Caroline Smith on research methods for measuring sweat rates over different parts of the body and the findings of this research.

ACTIVITY 12.2 *Exercise in the Heat* reviews changes in physiological responses due to exercising in the heat.

Health Risks During Exercise in the Heat

Measuring Heat Stress

Heat-Related Disorders

Sickle Cell Trait Complications

Preventing Hyperthermia

ACTIVITY 12.3 *Heat-Related Disorders* investigates three athlete scenarios covering the signs, symptoms, and treatment of heat cramps, heat exhaustion, and heatstroke.

Acclimation to Exercise in the Heat

Effects of Heat Acclimation
Achieving Heat Acclimation
Sex Differences

Exercise in the Cold

Habituation and Acclimation to Cold
Other Factors Affecting Body Heat Loss
Heat Loss in Cold Water

Physiological Responses to Exercise in the Cold

Muscle Function
Metabolic Responses

ACTIVITY 12.4 *Exercise in the Cold* reviews changes in physiological responses due to exercising in the cold.

Health Risks During Exercise in the Cold

Hypothermia
Cardiorespiratory Effects
Frostbite
Exercise-Induced Asthma

ANIMATION FOR FIGURE 12.15 shows the warming of air as it enters the respiratory system.

ACTIVITY 12.5 *Cold-Related Health Risks* investigates two recreational scenarios covering the signs, symptoms, and treatment of health risks related to exercising in the cold.

In Closing



Organizers of the 2014 Australian (Tennis) Open were criticized for forcing players to compete in intense heat as temperatures hit 42 °C (108 °F) in Melbourne on January 13. The day's peak temperature of 42 °C was just short of Melbourne's January record of 45.6 °C (114 °F), which occurred in 1936. The Olympic Movement Medical Code states, "In each sports discipline, minimal safety requirements should be defined and applied with a view to protecting the health of the participants and the public during training and competition. Depending on the sport and the level of competition, specific rules should be adopted regarding sports venues [and] safe environmental conditions."⁵ All major sporting bodies abide by this code and have comprehensive management strategies in place. However, the Australian Open Extreme Heat Policy (EHP) is applied only at the referee's discretion, and only minimal changes were enacted to protect the players in Melbourne.

Several prominent competitors, including Scotland's Andy Murray, unsuccessfully called on Australian Open organizers to reconsider their decision to make players compete in such oppressive temperatures. Canadian player Frank Dancevic passed out during the second set of his first-round match with France's Benoit Paire on an unshaded court, as did a ball boy. It was so hot that Danish player Caroline Wozniacki's plastic water bottle melted on court and Serbia's Jelena Jankovic burned her backside and hamstrings sitting on an uncovered seat, then fell during her first-round victory when her rubber-soled shoe stuck to the court. Yet play continued.

The stresses of physical exertion are often complicated by environmental conditions. Performing exercise in extreme heat or cold places an additional burden on the mechanisms that regulate body temperature while supporting continued exercise. Although these mechanisms are amazingly effective in regulating body temperature under normal conditions, mechanisms of **thermoregulation** can be inadequate when we are subjected to extreme heat or cold. Fortunately, the body is able to adapt to such environmental stresses with continued exposure over time, a process known as *acclimation* (which refers to a short-term adaptation, e.g., days to weeks) or *acclimatization* (the proper term when we are referring to natural adaptations gained over long periods of time, e.g., months to years).

In the following discussion, we focus on the physiological responses to acute and chronic exercise in both hot and cold environments. Specific health risks are associated with exercise in both temperature extremes, so we also discuss the prevention of temperature-related illness and injuries during exercise.

Body Temperature Regulation

Humans are homeotherms, which means that our internal body temperature is physiologically regulated to keep it nearly constant even when environmental temperature changes. In physiology, temperatures are expressed as degrees Centigrade. To convert from °F to °C and vice versa, use the following transformations:

To go from °F to °C: Subtract 32°, then divide by 1.8.

To go from °C to °F: Multiply by 1.8, then add 32°.

Although a person's temperature varies from day to day, and even from hour to hour, these fluctuations are usually no more than about 1.0 °C (1.8 °F). Only during prolonged heavy exercise, fever due to illness, or extreme conditions of heat or cold do body temperatures deviate from the normal baseline range of 36.1 to 37.8 °C (97.0-100.0 °F). Body temperature reflects a careful balance between heat production and heat loss. Whenever this balance is disturbed, body temperature changes.

Metabolic Heat Production

Only a small part (usually less than 25%) of the energy (adenosine triphosphate, ATP) the body produces is used for physiological functions such as muscle contraction; the rest is converted to heat. All active tissues produce metabolic heat (M) that must be precisely offset by heat loss to the environment to maintain the internal temperature of the body. If the body's heat production exceeds its heat loss, as it often does during moderate- to heavy-intensity aerobic activity, the body stores the excess heat, and internal temperature increases. People's ability to maintain a constant internal temperature depends on their ability to balance the metabolic heat they produce and the heat they gain from the

environment with the heat the body loses. This balance is depicted in [figure 12.1](#).

Transfer of Body Heat to and From the Environment

Let's examine the mechanisms by which heat is transferred between a person and the surroundings. For the body to transfer heat to the environment, the heat produced in the body must first move from deep in the body (the core) to the skin (the shell), where it has access to the outside environment. Heat is primarily moved from the core to the skin by the blood. Only when heat reaches the skin can it be transferred to the environment by any of four mechanisms: conduction, convection, radiation, and evaporation. These are illustrated in [figure 12.2](#).

Conduction and Convection

Heat **conduction (K)** involves the transfer of heat from one solid material to another through direct molecular contact. As an example, heat can be lost from the body when the skin is in contact with a cold object, as when one sits on cold metal bleachers watching a soccer match on a chilly day. Conversely, if a hot object is pressed against the skin, heat from the object is conducted to the skin and heat is gained by the body. If the contact is prolonged, heat from the skin surface can be transferred to the blood as it flows through the skin and transferred to the core, raising internal (core) temperature. During exercise, conduction is usually negligible as a source of heat exchange because the body surface area in contact with solid objects (for example, soles of the feet on hot playing fields) is small. Therefore, many environmental physiologists treat conductive heat exchange as negligible in their calculations of heat balance and exchange.

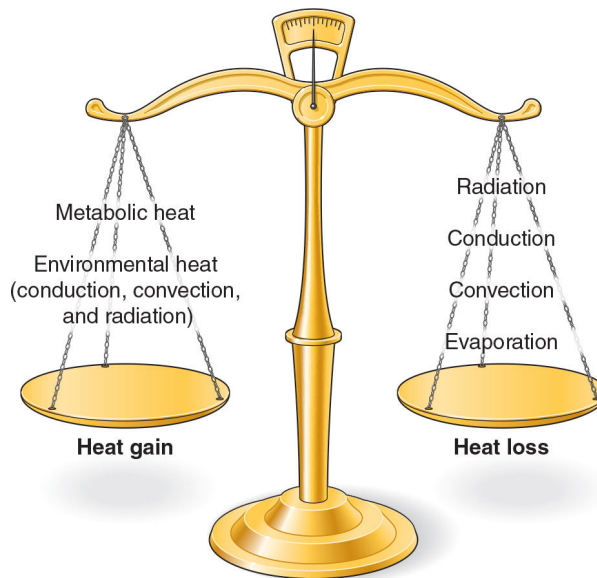


FIGURE 12.1 To maintain a steady-state core temperature, the body must balance the heat gained from metabolism and external environmental factors with the heat lost through the avenues of radiation, conduction, convection, and evaporation.

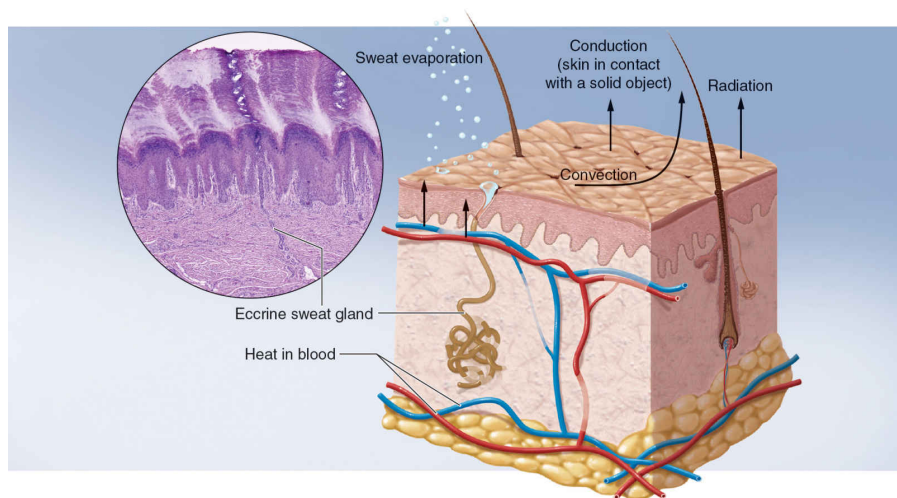


FIGURE 12.2 The removal of heat from the skin. Heat is delivered to the body surface via the arterial blood and to a lesser extent by conduction through the subcutaneous tissue. When the temperature of the skin is greater than that of the environment, heat is removed by conduction (if the skin is in contact with an object), convection, radiation, and sweat evaporation; when the environmental temperature exceeds skin temperature, heat can be removed only by evaporation.

Convection (C), on the other hand, involves transferring heat by the motion of a gas or a liquid across the heated surface. When the body is still and there is little air movement, a thin unstirred boundary layer of air surrounds the body. However, the air around us is usually in constant motion, especially so during exercise as we move either

the whole body or body segments (e.g., the arms pumping as we run) through the air. As air moves around us, passing over the skin, heat is exchanged with the air molecules. The greater the movement of the air (or liquid, such as water), the greater the rate of heat exchange by convection. Thus, in an environment in which air temperature is cooler than the skin temperature, convection permits the transfer of heat from the skin to the air (heat loss); however, if air temperature is higher than skin temperature, heat is gained by the body through convection. We often consider these processes to be mechanisms of heat loss, forgetting that when the environmental temperature exceeds skin temperature, the gradient works in the opposite direction.

Convection is important on a daily basis, since it constantly removes the metabolic heat we generate at rest and during activities of daily living, as long as the air temperature is lower than the skin temperature. However, if a person is submerged in cold water, the amount of heat dissipated from the body to the water by convection can be nearly 26 times greater than when the person is exposed to a similar cold air temperature.

Radiation

At rest, **radiation (R)** and convection are the primary methods for eliminating the body's excess heat. At normal room temperature (typically 21-25 °C, or ~70-77 °F), the nude body loses about 60% of its excess heat by radiation. The heat is given off in the form of infrared rays, which are a type of electromagnetic wave. [Figure 12.3](#) shows two infrared thermograms of an individual.

The skin constantly radiates heat in all directions to objects around it, such as clothing, furniture, and walls, but it also can receive radiant heat from surrounding objects that are warmer. If the temperature of the surrounding objects is greater than that of the skin, the body experiences a net heat gain via radiation. A tremendous amount of radiant heat is received from exposure to the sun.

Taken together, conduction, convection, and radiation are considered avenues of **dry heat exchange**. Resistance to dry heat exchange is called **insulation**, a concept known to everyone as it

relates to clothing and home heating and cooling. The ideal insulator is a layer of still air (remember that moving air causes convective heat loss), which we achieve by trapping layers of air in fibers (down, fiberglass, and so on). Adding insulation in this way minimizes unwanted heat loss in cold environments. However, during exercise we want to dissipate heat to the environment, which is best accomplished by wearing thin, light-colored clothing (to limit radiant heat absorption) that allows for a maximally exposed skin surface area. The effect of clothing on sweat evaporation is discussed later.

Evaporation

By far, **evaporation (E)** is the primary avenue for heat dissipation during exercise. In fact, when air temperature is close to skin temperature, the only available means of cooling is evaporation. As a fluid evaporates and turns into its gaseous form, heat is lost. An important concept to remember is that sweat must evaporate for any heat loss to occur. Some sweat drips off the body or stays on the skin or in the clothing, especially if the air is humid. This unevaporated sweat contributes nothing to body cooling and simply represents a wasteful loss of body water. Evaporation accounts for about 80% of the total heat loss when one is physically active and is therefore an extremely important avenue for heat loss. Even at rest, evaporation accounts for 10% to 20% of body heat loss, since some evaporation occurs without our awareness (termed *insensible water loss*).

As body core temperature increases, a threshold core temperature is reached, and production increases dramatically. As sweat reaches the skin, it is converted from a liquid to a vapor, and heat is lost from the skin in the process, the latent heat of vaporization. Thus, sweat evaporation becomes increasingly important as body temperature increases.

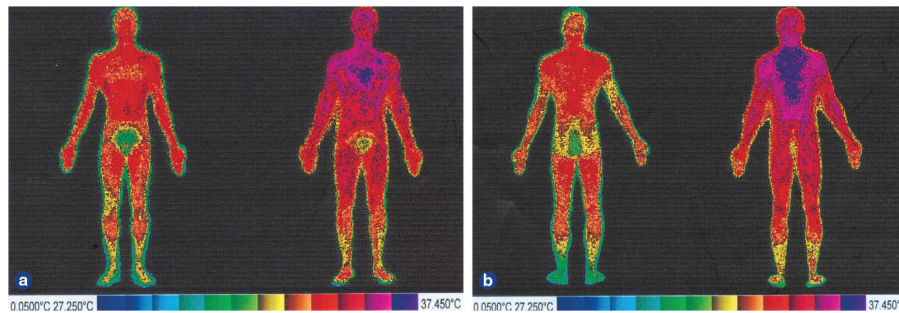


FIGURE 12.3 Thermograms of the body showing the variations in radiant (infrared) heat leaving the (a) front and (b) rear surfaces of the body both before (left) and after (right) running outside at a temperature of 30 °C and 75% humidity. The color scale at the bottom of each photo provides the temperature variation for the changes in color.

Evaporation of 1 L of sweat in an hour results in the loss of 680 W (2,428 kJ) of heat. To put the tremendous cooling power of sweat evaporation into perspective, a 70 kg (154 lb) marathon runner who runs a 2:30 marathon will produce about 1,000 W of metabolic heat. If every drop of sweat he produced evaporated, he would need to sweat at a rate of about 1.5 L per hour. Because inevitably some sweat drips off, a better estimated sweating rate necessary to maintain core temperature would be 2 L per hour. Without replacing fluids, however, our hypothetical runner would lose about 5 L (5.3 qt) of water, or over 7% of his body weight!

Analogous to insulation, which limits dry heat exchange, clothing also adds resistance to sweat evaporation. While some cooling of the skin does occur as sweat evaporates from wet clothing surfaces, the cooling power is less than for evaporation directly from skin to air. Clothing that fits loosely and comprises fabrics that promote wicking or free movement of water vapor molecules through the fabric enhances evaporative cooling.

Figure 12.4 shows the complex interaction between the mechanisms of body heat balance (production and loss) and environmental conditions.⁶ Using the symbols defined in the previous paragraphs, we can represent the state of heat balance by a simple equation:

$$M - W \pm R \pm C \pm K - E = 0,$$

where W represents any useful work being performed as a result of muscle contraction. Notice that while R, C, and K can be either

positive (heat gain) or negative (heat loss), E can only be negative. When $M - W \pm R \pm C \pm K - E > 0$, heat is stored in the body and core temperature rises.

Humidity and Heat Loss

The water vapor pressure of the air (the pressure exerted by water vapor molecules suspended in the air) plays a major role in evaporative heat loss. *Relative humidity* is a more commonly used term that relates the water vapor pressure of the air to that of fully saturated air (100% humidity). When humidity is high, the air already contains many water molecules. This decreases its capacity to accept more water because the vapor pressure gradient between the skin and the air is decreased. Thus, high humidity limits sweat evaporation and heat loss, while low humidity offers an ideal opportunity for sweat evaporation and heat loss. But this efficient cooling mechanism can also pose a problem. If sweating is prolonged without adequate fluid replacement, dehydration can occur.

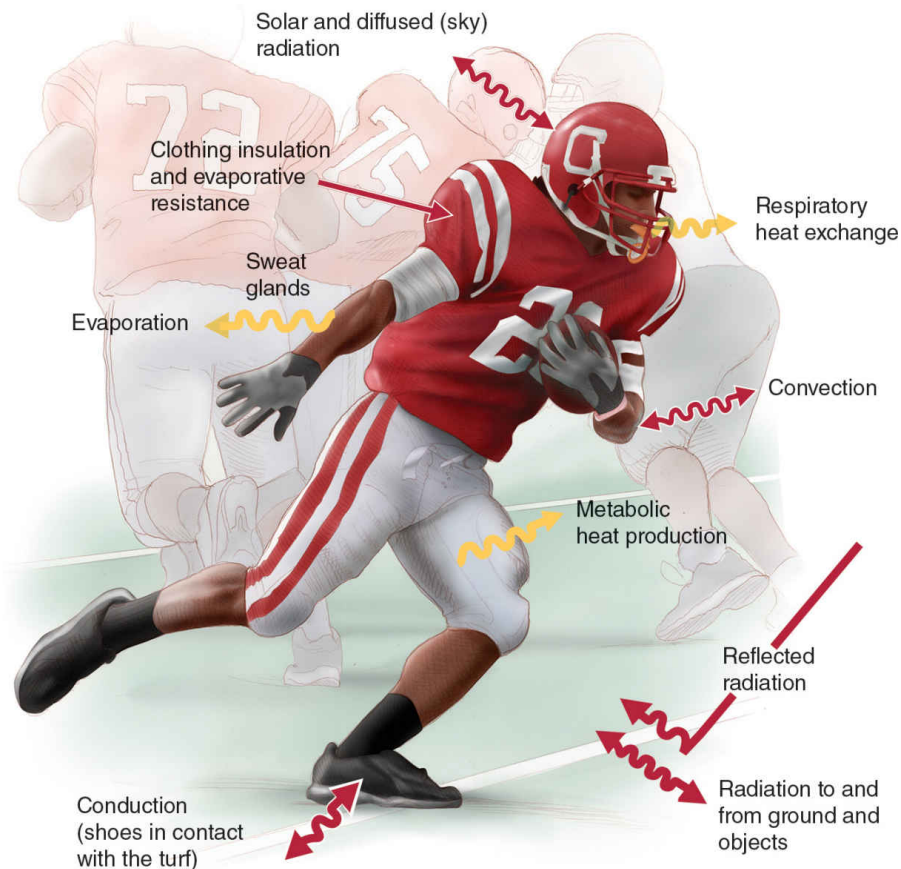


FIGURE 12.4 The complex interaction between the body's mechanisms for heat balance and environmental conditions.

Thermoregulatory Control

We live our entire lives within a very small, fiercely protected range of internal body temperatures. If sweating and evaporation were unlimited, we could withstand extreme ambient heat (e.g., even oven temperatures $>200\text{ }^{\circ}\text{C}$ for short periods!) if we were protected from contact with hot surfaces. On the other hand, the temperature limits for living cells range from about $0\text{ }^{\circ}\text{C}$ (where ice crystals form) to about $45\text{ }^{\circ}\text{C}$ (where intracellular proteins start to unravel), and humans can tolerate *internal* temperatures below $35\text{ }^{\circ}\text{C}$ or above $41\text{ }^{\circ}\text{C}$ for only very brief periods of time. To maintain internal temperature within these limits, we have developed very effective and, in some instances specialized, physiological responses to heat and cold. These responses involve the finely controlled coordination of several body systems.

Internal body temperature at rest is regulated at approximately 37 °C (98.6 °F). During exercise, the body is often unable to dissipate heat as rapidly as it is produced. In rare circumstances, a person can reach internal temperatures exceeding 40 °C (104 °F), with a temperature above 42 °C (107.6 °F) in active muscles. The muscles' energy systems become more chemically efficient with a small increase in muscle temperature, but internal body temperatures above 40 °C can adversely affect the nervous system and reduce further efforts to unload excess heat. How does the body regulate its internal temperature? The hypothalamus plays a central role (see [figure 12.5](#)).

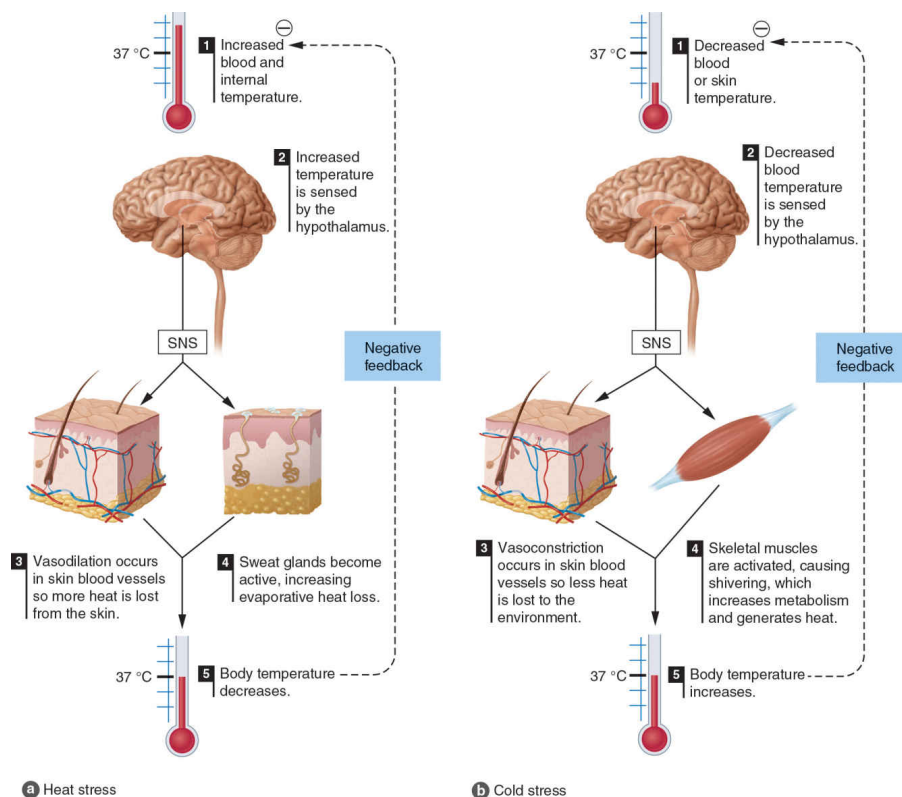


FIGURE 12.5 A simplified overview of the role of the hypothalamus in controlling body temperature during (a) heat stress and (b) cold stress.



The Preoptic-Anterior Hypothalamus: The Body's Thermostat

A simple way to envision the mechanisms that control internal body temperature is to compare them to the thermostat that controls the air temperature in a home, although the body's mechanisms function in a more complex manner and generally with greater precision than a home heating and cooling system. Sensory receptors called **thermoreceptors** detect changes in temperature and relay this information to the body's thermostat, located in a region of the brain called the **preoptic-anterior hypothalamus (POAH)**. In response, the hypothalamus activates mechanisms that regulate the heating or cooling of the body. Like a home thermostat, the hypothalamus has a predetermined temperature, or set point, that it tries to maintain. This is the normal body temperature. The smallest deviation from this set point signals this thermoregulatory center to readjust the body temperature.

Thermoreceptors are located throughout the body but especially in the skin and central nervous system. The peripheral receptors located in the skin monitor the skin temperature, which varies with changes in the temperature around a person. They provide information not only to the POAH but also to the cerebral cortex, which allows one to consciously perceive temperature and voluntarily control one's exposure to heat or cold. Because the skin temperature changes long before core temperature, these receptors serve as an early warning system for impending thermal challenges.

Central receptors are located in the hypothalamus, other brain regions, and the spinal cord and monitor the temperature of the blood as it circulates through these sensitive areas. These central

receptors are responsive to blood temperature changes as small as 0.01 °C (0.018 °F) and to the rate of change as well. Because of this exquisite sensitivity, very small changes in the temperature of the blood passing through the hypothalamus quickly trigger reflexes that help one conserve or eliminate body heat as needed.

Environmental challenges to thermal homeostasis present challenges in parallel to many other bodily control systems, such as those that control blood pressure, body fluid and electrolyte balance, and circadian rhythms. In testimony to the intricate design of the human brain, these other hypothalamus-based controllers are located in close proximity to the POAH, with neural connections that finely coordinate control across all of these systems.



Thermoregulatory Effectors

When the POAH senses temperatures above or below normal, signals are sent through the sympathetic nervous system to four sets of effectors:

- **Skin arterioles.** When skin or core temperature changes, the POAH sends signals via the sympathetic nervous system (SNS) to the smooth muscle in the walls of the arterioles that supply the skin, causing them to dilate or constrict. This either increases or decreases skin blood flow. Skin vasoconstriction results primarily from SNS release of the neurotransmitter norepinephrine, although other neurotransmitters are involved as well, and facilitates heat conservation by minimizing dry heat exchange. Skin vasodilation in response to heat stress is a more complex and less understood process. Increased skin blood flow aids in heat dissipation to the environment through conduction, convection, and radiation (and, indirectly, evaporation as skin temperature goes up). Fine-tuning of skin blood flow is the mechanism by which minute-to-minute adjustments are made in heat balance and exchange. These adjustments are rapid and occur with no real energy cost to the body.

- **Eccrine sweat glands.** When either the skin or core temperature is elevated sufficiently, the POAH also sends impulses through the SNS to the **eccrine sweat glands**, resulting in active secretion of sweat onto the skin surface. The primary neurotransmitter involved is acetylcholine; thus we refer to sweat gland activation as sympathetic cholinergic stimulation. Like skin arterioles, sweat glands are about 10 times more responsive to increases in core temperature than to similar increases in skin temperature. The evaporation of this moisture, as discussed earlier, removes heat from the skin surface.

- **Skeletal muscle.** Skeletal muscle is called into action when a person needs to generate more body heat. In a cold environment, thermoreceptors in the skin or core sense cold and send signals to the hypothalamus. In response to this integrated neural input, the hypothalamus activates the brain centers that control muscle tone. These centers stimulate shivering, which is

a rapid, involuntary cycle of contraction and relaxation of skeletal muscles. This increased muscle activity is ideal for generating heat to either maintain or increase the body temperature, because no useful work results from the shivering, only heat production.

- **Endocrine glands.** The effects of several hormones cause the cells to increase their metabolic rates. Increased metabolism affects heat balance because it increases heat production. Cooling the body stimulates the release of thyroxine from the thyroid gland. Thyroxine can elevate the metabolic rate throughout the body by more than 100%. Also, recall that epinephrine and norepinephrine (the catecholamines) mimic and enhance the activity of the SNS. Thus, they directly affect the metabolic rate of virtually all body cells.

In Review

- Humans are homeothermic, meaning that internal body temperature is regulated through physiological mechanisms, usually keeping it in the resting range of 36.1 to 37.8 °C (97.0-100.0 °F) despite changes in environmental temperatures.
- Body heat is transferred by conduction, convection, radiation, and evaporation. At rest, most heat is lost via radiation and convection, but during exercise, evaporation becomes the most important avenue of heat loss.
- When air temperature is close to skin temperature, the only available means of cooling is evaporation of sweat. Evaporation of 1 L of sweat in an hour results in the loss of 680 W (2,428 kJ) of heat.
- At any given air temperature, higher humidity (i.e., a higher water vapor pressure of the surrounding air) decreases evaporative heat loss.
- The preoptic-anterior area of the hypothalamus (POAH) is the primary thermoregulatory center. It acts as a thermostat, monitoring temperature and accelerating heat loss or heat production as needed.
- Two sets of thermoreceptors provide temperature information to the POAH. Peripheral receptors in the skin relay information about the temperature of the skin and the environment around it. Central receptors in the hypothalamus, other brain areas, and the spinal cord transmit information about the internal body temperature. Central thermoreceptors are far more sensitive to temperature

change than peripheral receptors. The main role of the peripheral receptors is anticipatory, allowing for early adjustments to be made.

- Effectors stimulated by the hypothalamus through the SNS can alter the body temperature. Increased skeletal muscle activity (voluntary or involuntary as in the case of shivering) increases the temperature by increasing metabolic heat production. Increased sweat gland activity decreases the temperature by increasing evaporative heat loss. Smooth muscle in the skin arterioles can cause these vessels to dilate to direct blood to the skin for heat transfer or to constrict to retain heat deep in the body. Metabolic heat production can also be stimulated by the actions of hormones such as thyroxine and the catecholamines.
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Physiological Responses to Exercise in the Heat

Heat production is beneficial during exercise in a cold environment because it helps maintain normal body temperature. However, even when exercise is performed in a cool environment, the metabolic heat load places a considerable burden on the mechanisms that control body temperature. In this section, we examine some physiological changes that occur in response to exercise while the body is exposed to heat stress and the impact that these changes can have on performance. For this discussion, heat stress means any environmental condition that increases body temperature and jeopardizes homeostasis.

Cardiovascular Function

As we learned in [chapter 8](#), exercise increases the demands on the cardiovascular system. When the need to regulate body temperature is added during exercise in the heat, the burden placed on the cardiovascular system is enhanced. During exercise in hot conditions, the circulatory system has to continue to transport blood not only to working muscle but also to the skin, where the tremendous heat generated by the muscles can be transferred to the environment. To meet this dual demand during exercise in the heat, two changes occur. First, cardiac output increases further (above the level associated with a similar exercise intensity in cool conditions) by increasing both heart rate and contractility. Second, blood flow is

shunted away from nonessential areas (like the gut, liver, and kidneys) and to the skin.

Consider what happens during prolonged running on a hot day. The aerobic exercise increases both metabolic heat production and the demand for blood flow and oxygen delivery to the working muscles. This excess heat can be dissipated only if blood flow increases to the skin.

In response to the elevated core temperature (and to a lesser extent, the higher skin temperature), the SNS signals sent from the POAH to the skin arterioles cause these blood vessels to dilate, delivering more metabolic heat to the body surface. Sympathetic nervous system signals also go to the heart to increase heart rate and cause the left ventricle to pump more forcefully. However, the ability to increase stroke volume is limited as blood pools in the periphery and less returns to the left atrium. To maintain cardiac output under such circumstances, the heart rate gradually creeps upward to help compensate for the decrease in stroke volume.

The causes of this phenomenon, known as cardiovascular drift, were discussed in [chapter 8](#). Because blood volume stays constant or even decreases (as fluid is lost in sweat), another phase of cardiovascular adjustment occurs simultaneously. Sympathetic nerve signals to the kidneys, liver, and intestines cause vasoconstriction of those regional circulations, which allows more of the available cardiac output to reach the skin without compromising muscle blood flow.

RESEARCH PERSPECTIVE 12.1

Dehydration Challenges the Cardiovascular System During Exercise in the Heat

Dehydration commonly occurs when people exercise for prolonged periods of time without adequate fluid replacement. This fall in total body fluid causes a reduction in sweat rate and skin blood flow, resulting in significant increases in body heat storage, hyperthermia, and physiological strain. A compromised blood flow to the active skeletal muscle has been proposed as an important factor in the chain of events that lead to fatigue during exercise in the heat, and the overall fall in total body fluid poses a significant challenge to thermoregulation and cardiovascular control during exercise.

Dehydration and the resulting increases in core temperature that occur during prolonged submaximal exercise in the heat (e.g., distance running, cycling) lead to increases in heart rate and total peripheral resistance, small reductions in blood volume and mean arterial pressure, and large reductions in stroke volume and cardiac output. These changes lead to marked reductions in blood flow to active and nonactive muscles, skin, and cerebral circulations. Importantly, these exercise-induced alterations in cardiovascular function and blood flow during dehydration are prevented with oral rehydration, or when dehydrated individuals exercise in the cold, preventing hyperthermia. Thus, the impact of dehydration on cardiovascular function and performance occurs only when dehydration is coupled with hyperthermia during prolonged exercise.

A progressive fall in cardiac output and stroke volume and a compensatory, although possibly detrimental, rise in heart rate are common features of the dehydration-induced cardiovascular strain observed during prolonged aerobic exercise in the heat. Progressive dehydration reduces total blood volume and venous return; these factors, combined with a reduced diastolic filling time (due to elevated heart rate), compound to reduce stroke volume. Importantly, dehydration and hyperthermia do not influence the contractile properties of the myocardium, and it is the reduction in filling during diastole that is primarily responsible for the lower cardiac output observed with dehydration. This reduction in cardiac output is, in turn, responsible for reductions in perfusion pressure and limb blood flow. It is unlikely that changes in mean arterial pressure are responsible for the reductions in cerebral blood flow. Recent research suggests that these reductions may be explained by hyperventilatory reductions in the partial pressure of CO₂; however, future studies will need to provide more evidence before these mechanisms are clear.

What Limits Exercise in the Heat?

Seldom are records set in endurance events, such as distance running, when the environmental heat stress is great. The factors that cause early fatigue when heat stress is superimposed on prolonged exercise have been the topic of some debate, and several theories have been proposed. None of these theories captures every situation but taken together they demonstrate the multiple control systems at work during thermoregulation.

At some point, the cardiovascular system can no longer compensate for the increasing demands of continuing endurance exercise and efficiently regulating the body's heat. Consequently, any factor that tends to overload the cardiovascular system or

interfere with heat dissipation can drastically impair performance, increase the risk of overheating, or both. Exercise in the heat becomes limited when heart rate approaches maximum, especially in untrained or nonheat-acclimated exercisers, as illustrated in [figure 12.6](#). Interestingly, working muscle blood flow is well maintained even at very high core temperatures unless significant dehydration occurs.

Another theory that helps explain limitations to exercise in the heat—especially in well-trained, acclimated athletes—is the **critical temperature theory**. This theory proposes that, regardless of the rate at which core temperature (and thus brain temperature) increases, the brain will send signals to stop exercise when some critical brain temperature is reached, usually between 40 and 41 °C (104 and 105.8 °F).

Precooling and Sport Performance

Many athletes now artificially lower their body core temperature before exercise, a practice known as *precooling*. Various methods are used, including cold-water immersion, sitting in a cold room, cold showers, cooling packs or vests, and ingesting a cold drink or ice slurry. If an athlete starts at a lower core temperature and his or her core temperature subsequently rises at the same rate, his or her core temperature will be lower at any given point in the exercise bout or competition. In laboratory studies, this often allows subjects to exercise longer at a given intensity under hot conditions. But does this practice improve sport performance?

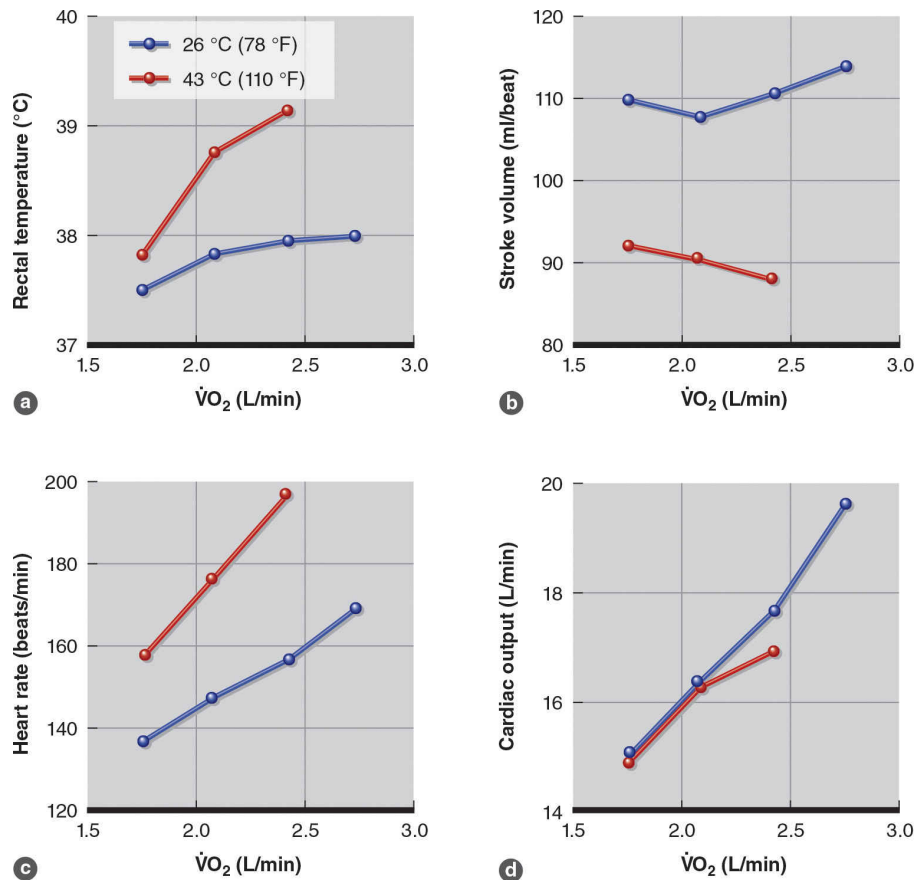


FIGURE 12.6 (a) Rectal temperature and (b-d) cardiovascular responses to graded exercise in thermoneutral (26 °C, or 78 °F; blue circles) and hot (43 °C, or 110 °F; red circles) environments. Note that, in addition to directional changes in these variables caused by heat stress, the maximal intensity is decreased when exercise is performed in a hot environment.

Based on Rowell (1974).

Two recent meta-analyses (statistically based review of existing studies) addressed that important question,^{14,15} concurring that precooling *does* effectively enhance sport performance, particularly in hot environments. As expected, precooling had a larger effect during endurance events like cycling time trials, but improvement was also seen for intermittent running or cycling sprints. The effectiveness of precooling on short sprints was minimal. Of the methods compared, ingesting cold drinks was the most promising procedure. Further, the best results occurred in athletes with the highest $\dot{V}O_{2\max}$.¹²

The second analysis¹¹ concluded that precooling actually impaired sprint performance, which might be expected if active muscle temperature fell. Intermittent activity and endurance performance

were both improved with precooling, however. Similarly, cooling maneuvers used during exercise were also successful at improving performance. Whether used before exercise or during exercise, the effectiveness of body cooling procedures depends on both (1) the amount of cooling (intensity and duration) and (2) how much thermal strain is involved in the exercise.

Body Fluid Balance: Sweating

On hot summer days, it is not uncommon for the temperature of the environment to exceed both the skin and deep body temperatures. As mentioned earlier, this makes evaporation far more important for heat loss, because radiation, convection, and conduction now become avenues of heat gain from the environment. Increased dependence on evaporation means an increased demand for sweating.

The eccrine sweat glands are controlled by stimulation of the POAH. Elevated blood temperature causes this region of the hypothalamus to transmit impulses through the sympathetic nerve fibers to the millions of eccrine sweat glands distributed over the body's surface. The sweat glands are fairly simple tubular structures extending through the dermis and epidermis, opening onto the skin, as illustrated in [figure 12.7](#).

A second type of sweat gland, the apocrine gland, is clustered in particular regions of the body, including the face, axilla, and genital regions. These are the glands associated with nervous perspiration, and they do not contribute significantly to heat loss by evaporation. The eccrine sweat glands, on the other hand, play a purely thermoregulatory role. They are located over most of the skin surface, with ~2 to 5 million covering the whole body. They are most densely distributed on the palms of the hands, the soles of the feet, and the forehead. The lowest densities are found on the forearms, lower legs, and thighs. When sweating begins, there are large regional variations in sweating rate. During exercise, the highest local sweating rates are typically measured on the middle and lower back and on the forehead, while the lowest sweating rates are observed on the hands and feet.

Sweat is formed in the coiled secretory portion of the sweat gland and at this stage has an electrolyte composition similar to that of the blood, since plasma is the source of sweat formation. As this filtrate of plasma passes through the uncoiled duct of the gland, sodium and chloride are reabsorbed back into the surrounding tissues and then into the blood. As a result, the final sweat that is extruded onto the skin surface through the sweat gland pores is hypotonic to (has less electrolytes than) plasma. During light sweating, the filtrate sweat travels slowly enough through the duct that there is time for reabsorption of sodium and chloride. Thus, the sweat that forms during light sweating contains very little of these electrolytes by the time it reaches the skin. However, when the sweating rate increases during exercise, the filtrate moves more quickly through the tubules, allowing less time for reabsorption, and the sodium and chloride content of sweat can be considerably higher.

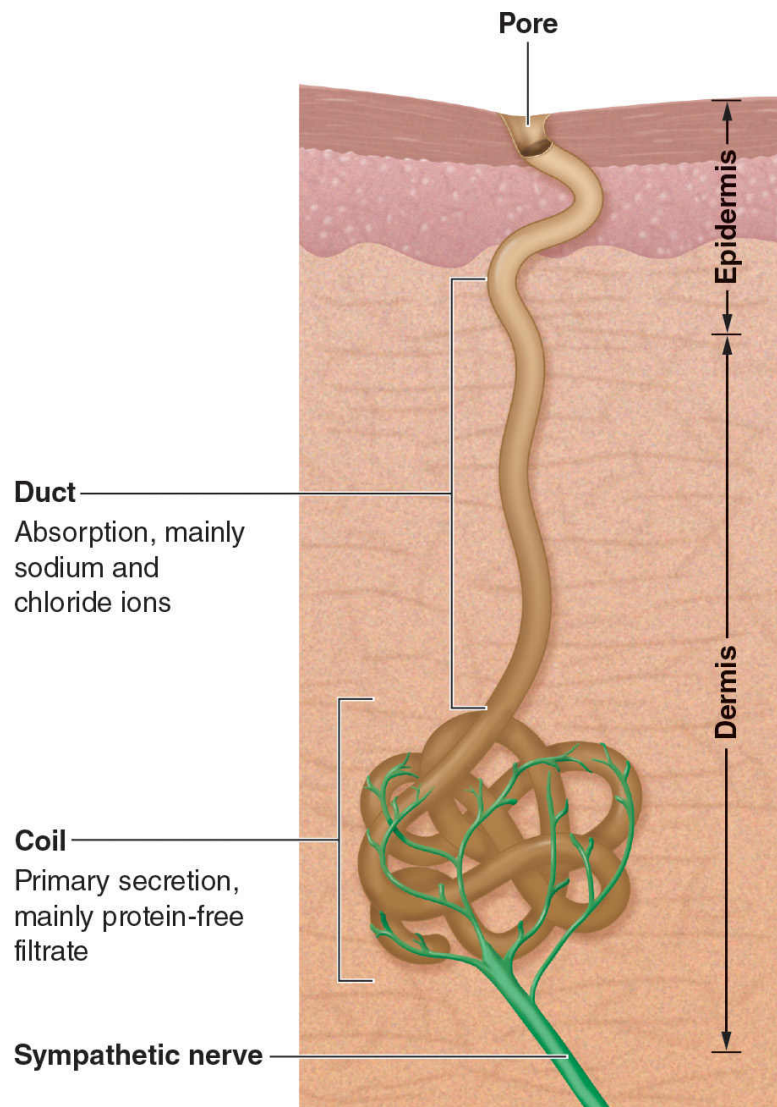


FIGURE 12.7 Anatomy of an eccrine sweat gland that is innervated by a sympathetic cholinergic nerve.

As seen in [table 12.1](#), the electrolyte concentration of trained and untrained subjects' sweat is significantly different. With training and repeated heat exposure (acclimation), more sodium is reabsorbed and the sweat is more dilute, in part because the sweat glands become more sensitive to the hormone aldosterone. Unfortunately, the sweat glands apparently do not have a similar mechanism for conserving other electrolytes. Potassium, calcium, and magnesium are not reabsorbed by the sweat glands and are therefore found in similar concentrations in sweat and plasma. In addition to heat

acclimation and aerobic training, genetics is a major determinant of both sweating rate and sweat sodium losses.

While performing heavy exercise in hot conditions, the body can lose more than 1 L of sweat per hour per square meter of body surface. This means that during intense effort on a hot and humid day (high level of heat stress), an average-sized female athlete (50-75 kg, or 110-165 lb) might lose 1.6 to 2.0 L of sweat, or about 2.5% to 3.2% of body weight, each hour. A person can lose a critical amount of body water in only a few hours of exercise in these conditions.

VIDEO 12.1 Presents Caroline Smith on research methods for measuring sweat rates over different parts of the body and the findings of this research.



A high rate of sweating maintained for a prolonged time ultimately reduces blood volume. This limits the volume of blood returning to the heart, increasing heart rate and eventually decreasing cardiac output, which in turn reduces performance potential, particularly for endurance activities. In long-distance runners, sweat losses can approach 6% to 10% of body weight. Such severe dehydration can limit subsequent sweating and make the individual susceptible to heat-related illnesses. [Chapter 15](#) provides a detailed discussion of dehydration and the value of fluid replacement.

Loss of both electrolytes and water in the sweat triggers the release of both aldosterone and antidiuretic hormone (ADH), also known as **vasopressin** or **arginine vasopressin**. Recall that

aldosterone is responsible for maintaining appropriate sodium concentrations in the blood and that ADH plays a key role in maintaining fluid balance (chapter 4). Aldosterone is released from the adrenal cortex in response to stimuli such as decreased blood sodium content, reduced blood volume, or reduced blood pressure. During acute exercise in the heat and during repeated days of exercise in the heat, this hormone limits sodium excretion from the kidneys. More sodium is retained by the body, which in turn promotes water retention. This allows the body to retain water and sodium in preparation for additional exposures to the heat and subsequent sweat losses.

Similarly, exercise and body water loss stimulate the posterior pituitary gland to release ADH. This hormone stimulates water reabsorption from the kidneys, which further promotes fluid retention in the body. Thus, the body attempts to compensate for loss of electrolytes and water during periods of heat stress and heavy sweating by reducing their loss in urine. Also, recall that blood flow to the kidneys is substantially reduced during exercise in the heat, which likewise aids in fluid retention.

TABLE 12.1 Example of Sodium, Chloride, and Potassium Concentrations in the Sweat of Trained and Untrained Subjects During Exercise

Subjects	Sweat Na (mmol/L)	Sweat Cl ⁻ (mmol/L)	Sweat K ⁺ (mmol/L)
Untrained men	90	60	4
Trained men	35	30	4
Untrained women	105	98	4
Trained women	62	47	4

Individual sweat electrolyte concentrations are highly variable, but training and heat acclimation decrease sodium losses in the sweat.

Data from the Human Performance Laboratory, Ball State University.

In Review

- During exercise in the heat, the skin competes with the active muscles for a limited cardiac output. Muscle blood flow is well maintained (sometimes to the detriment of skin blood flow) unless severe dehydration occurs. A series of cardiovascular adjustments shunt blood away from nonessential regions such as the liver, gut, and kidneys to the skin to aid in heat dissipation.